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EXP9:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char mat[5][5];

void build(const char *key) {
    int used[26] = {0}, r = 0, c = 0, i;
    char all[60];
    sprintf(all, "%sABCDEFGHIKLMNOPQRSTUVWXYZ", key);   /* J is merged with I */
    for (i = 0; all[i]; i++) {
        int ch = toupper(all[i]);
        if (ch == 'J') ch = 'I';
        if (!isalpha(ch) || used[ch - 'A']) continue;
        used[ch - 'A'] = 1;
        mat[r][c++] = ch;
        if (c == 5) { c = 0; r++; }
    }
}

void find(char ch, int *r, int *c) {
    int i, j;
    if (ch == 'J') ch = 'I';
    for (i = 0; i < 5; i++)
        for (j = 0; j < 5; j++)
            if (mat[i][j] == ch) { *r = i; *c = j; return; }
}
```

```
int main() {
    char key[60], line[400], ct[400];
    int n = 0, i, j;

    printf("Enter key: ");
    fgets(key, sizeof key, stdin);
    printf("Enter Playfair ciphertext: ");
    fgets(line, sizeof line, stdin);
    for (i = 0; line[i]; i++) if (isalpha(line[i])) ct[n++] = toupper(line[i]);
    ct[n] = '\0';
    key[strcspn(key, "\n")] = 0;

    build(key);
    printf("\nPlayfair matrix:\n");
    for (i = 0; i < 5; i++) {
        for (j = 0; j < 5; j++) printf("%c ", mat[i][j]);
        printf("\n");
    }

    printf("\nDecrypted text:\n");
    for (i = 0; i + 1 < n; i += 2) {
        int r1, c1, r2, c2;
        find(ct[i], &r1, &c1);
        find(ct[i + 1], &r2, &c2);
        if (r1 == r2)      { putchar(mat[r1][(c1 + 4) % 5]); putchar(mat[r2][(c2 + 4) % 5]); }
        else if (c1 == c2) { putchar(mat[(r1 + 4) % 5][c1]); putchar(mat[(r2 + 4) % 5][c2]); }
        else               { putchar(mat[r1][c2]); putchar(mat[r2][c1]); }
    }
    printf("\n");
    return 0;
```

}

OUTPUT:

```
Enter key: ROYALNEWZEALANDNAVY
Enter Playfair ciphertext: KXJEY UREBE ZWEHE WRYTU HEYFS KREHE GOYFI WTTTU OLKSY CAJPO BOTEI ZO
NTX BYBNT GONEY CUZWR GDSON SXBOU YWRHE BAAHY USEDQ

Playfair matrix:
R O Y A L
N E W Z D
V B C F G
H I K M P
Q S T U X

Decrypted text:
PTBOATONEOWENINELOSTINACTIONINBLACKESSSTRAITTWOMILESSWMERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMA
TIONX
```